Quiz Bowl RL Buzzer

Multiple-Choice Strategic Buzzing Under Incremental Clues

CS234 Final Project

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Problem

- Quiz bowl questions reveal evidence incrementally
- A good system must decide WHEN to buzz, not just WHAT to pick
- Buzz too early: higher risk of a wrong answer
- Buzz too late: lower strategic value and less chance to beat an opponent
- We study this in a multiple-choice setting so the answer space is controlled and evaluation is reproducible

Sequential decision problem

- Model quiz bowl as sequential decision over partial clues
- Question prefixes over time
- K = 4 answer options
- One correct option plus three distractors

Two policy families

Belief-feature buzzers

- Threshold, softmax-profile, and Bayesian baselines
- PPO on structured observations

T5 text-policy buzzers

- End-to-end supervised warm start plus PPO

Why Multiple Choice?

Advantages

- Eliminates aliasing and grading complexity
- Isolates the buzzing decision itself
- Makes evaluation controlled and reproducible

Challenge

- Naive distractors create artifacts
- Answer generation quality still matters

Design goals

- Keep the answer space constrained
- Make options hard enough that the agent must use clues
- Use anti-artifact guards: alias collision, token overlap, length ratio, and question-text overlap checks

Why Multiple Choice?

1. Load tossups
2. Build answer profiles
3. Construct MC questions
4. Score with a likelihood model
5. Convert beliefs into observations
6. Run buzzer agents
7. Evaluate

Method Overview

Our Contribution

We study pyramidal quiz bowl under a restricted multiple-choice action space with reinforcement learning.

- Multiple-choice POMDP formulation for quiz bowl
- PPO to learn when to wait versus buzz and answer
- Artifact-resistant MC construction with several distractor strategies
- Direct tests of whether learned policies use clues or answer-choice patterns
- Calibration-focused evaluation with ECE, Brier, and S_q beyond raw accuracy

Our Contribution

Best baseline by mean S_q: always_final

Metric	Best baseline	PPO	Reading
Mean S_q	0.386	0.326	higher better
Buzz accuracy	38.6%	34.1%	higher better
Mean buzz step	4.05	0.00	timing profile
Reward-like	0.080	-0.476	higher better
ECE	0.000	0.099	lower better
Brier	0.000	0.013	lower better

Use this as a smoke-test snapshot, not a final leaderboard: compare S_q, calibration, and timing together.

How To Read PPO Metrics

Why raw accuracy can dip

- PPO learns a timing policy, not just a final guess
- Waiting can avoid low-confidence answers
- That may lower raw buzz accuracy on some slices
- Reward design and training stability matter a lot
- So accuracy alone is not the right summary statistic

Current smoke: always_final buzz acc 38.6% vs PPO 34.1%

What to compare alongside it

- Mean S_q or task reward
- Mean buzz step / timing profile
- Calibration at buzz time
- Controls that test clue use versus answer-artifact use
- Whether the policy abstains or times out strategically

A PPO buzzer is only better if its timing behavior improves the metrics you actually care about.

How To Read PPO Metrics

Quiz bowl MC "buzzing" is optimal stopping with improving evidence

Markov tossup

Original tossup (verbatim)

QBReader ▫ 2022 ARCADIA ▫ packet 08 ▫ tossup 11

Given a set of nodes named for this person, a node is conditionally independent from the rest of a Bayesian network. The Baum-Welch algorithm is used to train a type of model named for this person that is used for multiple sequence alignment. The initial "burn-in" states of a process named for this person are discarded in methods like Gibbs sampling. The Metropolis-Hastings algorithm approximates an unknown distribution as the (*) stationary distribution of a process named for this person. Monte Carlo methods often use processes named for this person that have stochastic transition matrices. Dynamic programming is used to decode "hidden" models named for this person. The next state of a random process named for this person depends only on the current state. For 10 points, what Russian mathematician names a type of memoryless "chain?"

Accepts: Markov blankets; hidden Markov models; Markov chains

Clue progression
Early clues are obscure, middle clues narrow the field, and the final "memoryless chain" clue is a giveaway.

RL intuition
Those repeated anaphoric phrases all resolve to Markov, and the strategic decision is when to "buzz."

Presentation multiple-choice setting

The answer options are fixed from the start, using a cleaner mathematician-only set:

- A. Andrey Andreyevich Markov
- B. Leonhard Euler
- C. Carl Friedrich Gauss
- D. Augustin-Louis Cauchy

Decision rule
At each "prefix," should the policy "WAIT" or "BUZZ" on one listed answer?

Section 1: the posterior sharpens as "clues" arrive

first watch uncertainty collapse

As more "clues" arrive, the Markov posterior collapses onto one option

"prefix t": only the clues seen so far are visible

Only the opening referent is visible: "nodes named for this person..."

WAIT

Keep "waiting": Markov only leads narrowly over the other mathematicians, so another clue can still change the ranking.

With a fixed answer set, the policy is not discovering new options. It is deciding when the current posterior over Markov vs. the three distractors is sharp enough to "act."

Posterior Sharpening

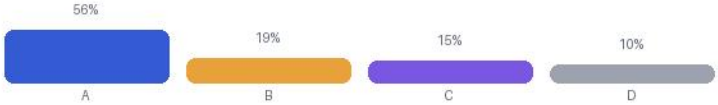
As more "clues" arrive, the Markov posterior collapses onto one option

"prefix t": only the clues seen so far are visible

The referent repeats: "a type of model named for this person..."

WAIT

Keep "waiting": Markov now leads, but continuation value is still meaningful.



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Posterior Sharpening

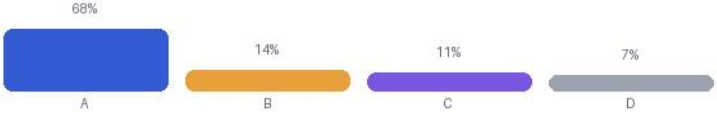
As more "clues" arrive, the Markov posterior collapses onto one option

"prefix t": only the clues seen so far are visible

Another repetition appears: "burn-in" states of a process named for this person..."

WAIT

Keep "waiting". A is favored, yet one more clue can still sharpen confidence.



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Posterior Sharpening

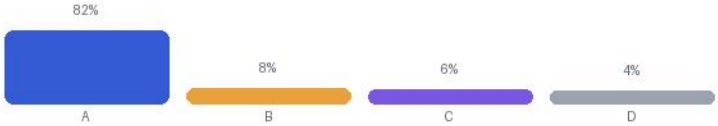
As more "clues" arrive, the Markov posterior collapses onto one option

"prefix t": only the clues seen so far are visible

The referent stays active: "stationary distribution of a process named for this person..."

BUZZ

"Buzz" now: the Markov reading is strong enough that waiting has less value.



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Posterior Sharpening

As more "clues" arrive, the Markov posterior collapses onto one option

"prefix t": only the clues seen so far are visible

Finally, "memoryless chain" turns the repeated "this person" referent into a giveaway.

BUZZ

"Buzz" now: the final clue mostly confirms what the posterior already says.

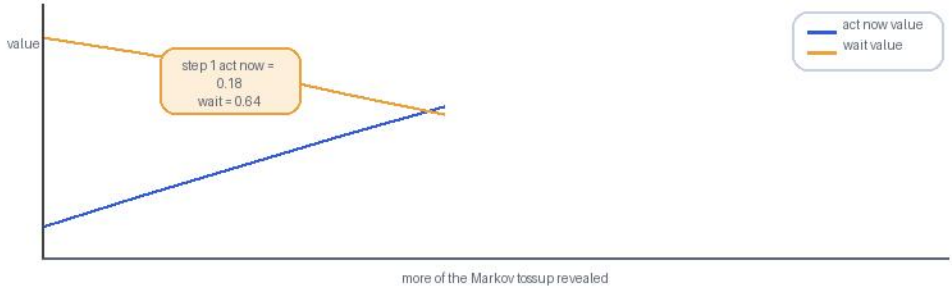
With a fixed answer set, the policy is not discovering new options. It is deciding when the current posterior over Markov vs. the three distractors is sharp enough to "act."

Posterior Sharpening

Section 2. "buzz" when acting now beats waiting

next: compare current value to one more clue

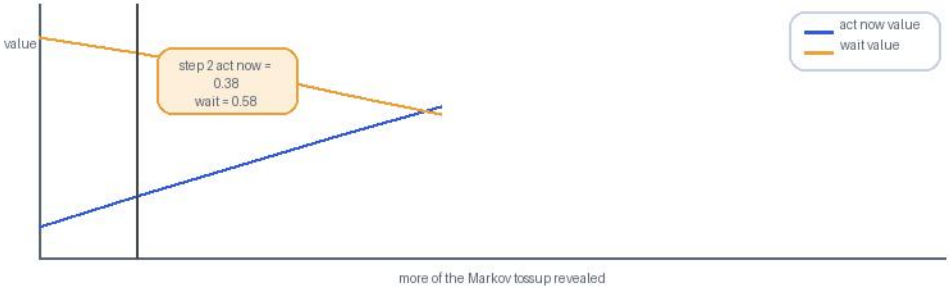
"Buzz" when acting now is worth more than waiting in the Markov example



On the first Markov clue, another "clue" is still worth much more than committing.

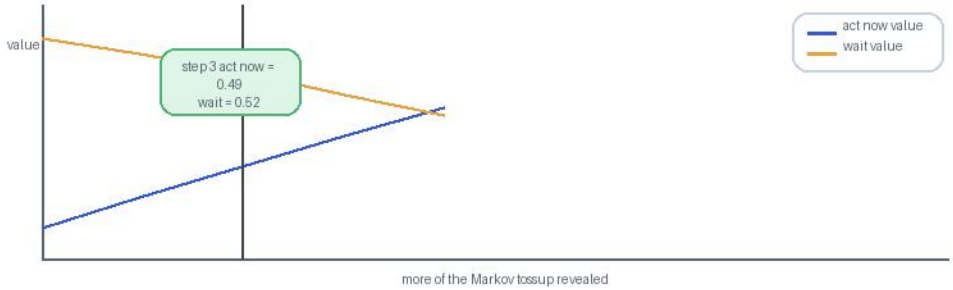
Act Now Vs Wait

"Buzz" when acting now is worth more than waiting in the Markov example



Baum-Welch pushes the example toward Markov, but waiting still wins.

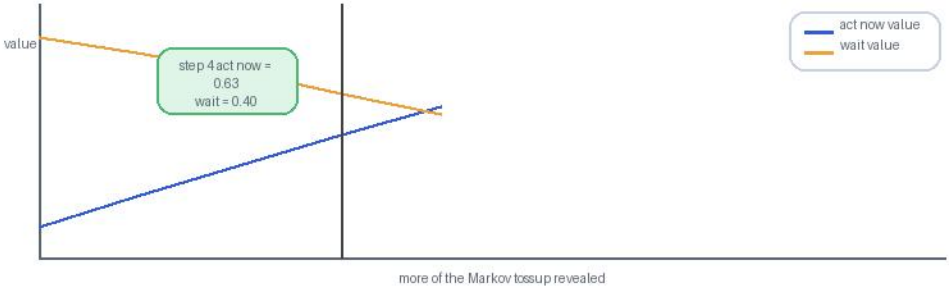
"Buzz" when acting now is worth more than waiting in the Markov example



Near the Gibbs / Metropolis region, timing becomes the whole game.

Act Now Vs Wait

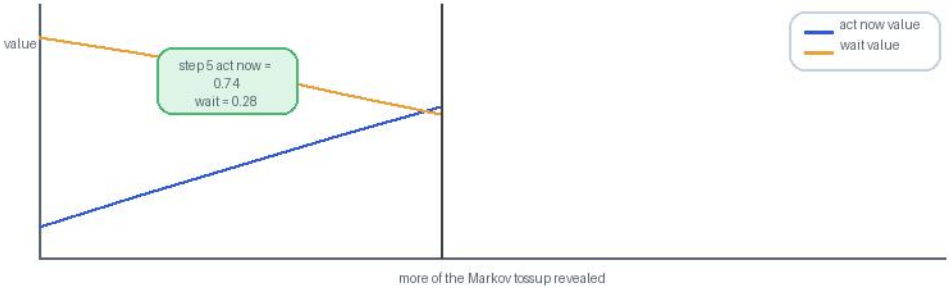
"Buzz" when acting now is worth more than waiting in the Markov example



Once the random-process clue arrives, "buzzing" overtakes waiting.

Act Now Vs Wait

"Buzz" when acting now is worth more than waiting in the Markov example



By the "memoryless chain" clue, waiting mostly delays a strong answer.

Act Now Vs Wait

Section 3: "ABSTAIN" means wait, not "None-of-the-Above"

next: waiting is about value of information

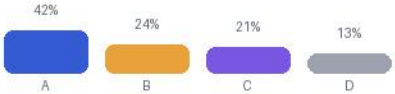
"ABSTAIN" means wait, not "none of the above"

The correct answer is already in the "set"

- A — Andrey Andreyevich Markov
- B — Leonhard Euler
- C — Carl Friedrich Gauss
- D — Augustin-Louis Cauchy

"Abstaining" means one more "clue" is worth more than committing, even though Markov is already option A.

What changes over "time" in the Markov example



ABSTAIN

means continue because another "clue" still has value

None-of-the-Above

Even with Markov already listed at A, "ABSTAIN" can still be rational after the earliest clue.

"ABSTAIN" means wait, not "none of the above"

The correct answer is already in the "set"

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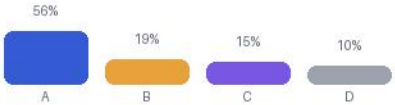

ABSTAIN

means continue because another "clue" still has value

None-of-the-Above

"Waiting" is still about the value of one more clue, not about missing options.

"ABSTAIN" means wait, not "none of the above"

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"Abstaining" means one more "clue" is worth more than committing, even though Markov is already option A.

What changes over "time" in the Markov example

ABSTAIN

means continue because another "clue" still has value

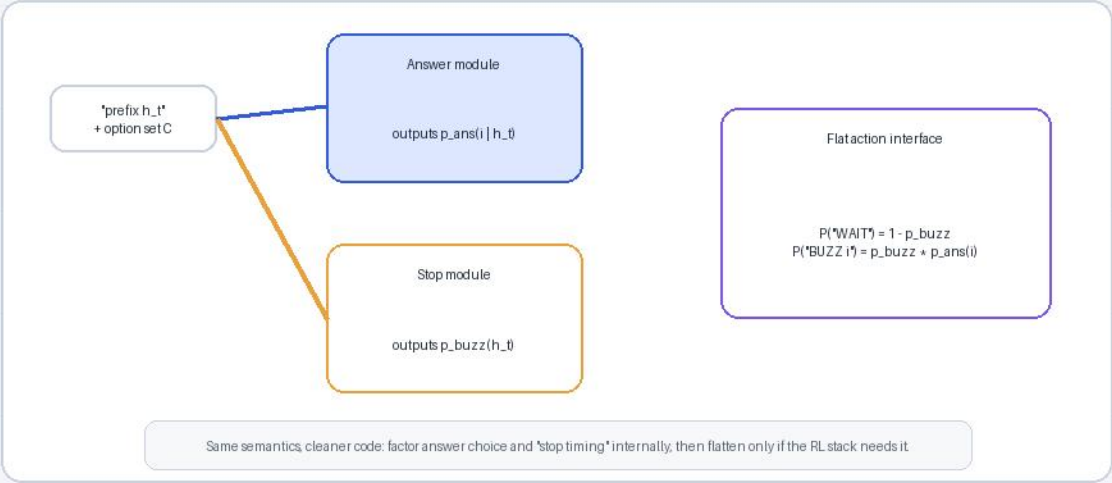
None-of-the-Above

So "ABSTAIN" means continuation value, even when the correct answer has been on the board all along.

Section 4: separate answer quality from "buzz" timing

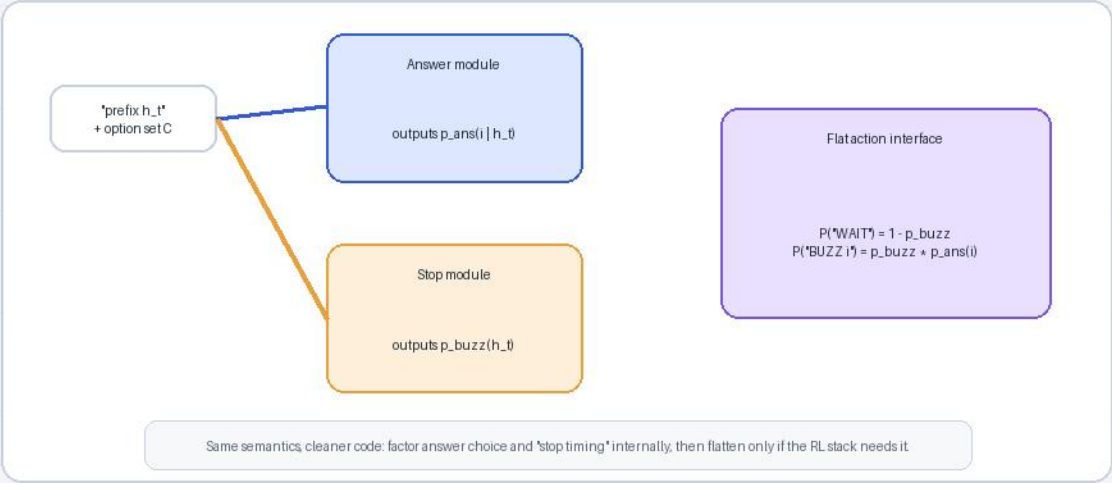
next: factor the decision into two modules

Implementation idea: separate answer quality from "buzz" timing



Factor Stop And Answer

Implementation idea: separate answer quality from "buzz" timing



Factor Stop And Answer

Section 5: one compact mental model

finally: compress the whole story

A practical training recipe

1. train the answer model

learn p_ans(i | h_t)

2. warm-start the stop head

learn when to "buzz"

3. PPO fine-tuning

optimize task reward

RL can then focus mostly on "timing" instead of learning answer choice and stopping from scratch at the same time.

Training setup in code

scripts/build_mc_dataset.py --smoke -> scripts/run_baselines.py --smoke -> scripts/train_ppo.py --smoke

Data + baseline prep

- build MC prefixes
- set distractor strategy
- run baseline sweeps

RL fine-tune

- initialize policy
- optimize timing reward
- trade speed versus accuracy

Artifacts

artifacts/smoke/
- mc_dataset.json
- baseline_summary.json
- ppo_runs.json

Evaluation protocol

Core metrics

- mean S_q
- buzz accuracy
- mean buzz step
- ECE and Brier

Control checks

- choices-only baseline
- shuffle test
- alias substitution
- per-category slices

Evaluation source of truth: artifacts/smoke/evaluation_report.json

Best baseline (mean S_q)

always_final

- mean S_q = 0.386
- buzz acc = 38.6%
- mean step = 4.05
- n = 44

PPO smoke

- mean S_q = 0.326
- buzz acc = 34.1%
- reward-like = -0.476
- ECE/Brier = 0.099/0.013

Full eval: S_q=0.243, accuracy=38.6%, mean step=3.50

Metrics

- S_q (QANTA system score)
- Buzz accuracy
- Mean buzz step
- Calibration-at-buzz / ECE / Brier
- Per-category accuracy

Controls

- Choices-only baseline (no question text)
- Shuffle clue order
- Alias substitution
- Distractor difficulty variation

A buzzer can look strong while exploiting answer artifacts instead of clue content.

Quiz bowl MC "buzzing," in one picture

1. the Markov "clues" go from Bayesian-network jargon to a giveaway

2. the posterior over the fixed four choices sharpens onto Markov

Mental model: selective prediction + optimal stopping over a fixed answer "set" that already contains Markov.

For RL people

At each "prefix," compare the value of acting now with the value of one more "clue."

Summary

Quiz bowl MC "buzzing," in one picture

1. the Markov "clues" go from Bayesian-network jargon to a giveaway
2. the posterior over the fixed four choices sharpens onto Markov
3. the key RL decision is when to stop "waiting" and buzz A

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At each "prefix," compare the value of acting now with the value of one more "clue."

If you keep a flat action interface, factor it internally:
P("WAIT")=1-p_buzz
P("BUZZ i")=p_buzz * p_ans(i)

Summary

Quiz bowl MC "buzzing," in one picture

1. the Markov "clues" go from Bayesian-network jargon to a giveaway
2. the posterior over the fixed four choices sharpens onto Markov
3. the key RL decision is when to stop "waiting" and buzz A
4. "abstain" means another clue is worth it, not "None-of-the-Above"

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Key takeaway: "pyramidal" quiz bowl turns multiple-choice QA into an optimal-stopping problem because evidence improves before action.

Summary

- Rodriguez et al. (2019) ▨ Quizbowl: The case for incremental QA
- Schulman et al. ▨ Proximal Policy Optimization (PPO)
- Raffel et al. ▨ T5: Exploring the Limits of Transfer Learning
- Boyd-Graber and Daume (2013) ▨ Bayesian thinking on your feet
- Boyd-Graber and Borschinger (2020) ▨ What QA can learn from trivia nerds
- Balepur et al. (2025) ▨ Test-time reasoners are strategic MC test-takers
- Kalai et al. (2025) ▨ Why language models hallucinate
- UMD / QANTA (2024) ▨ S_q evaluation metric